

George Youhana Heazil

📍 Cairo | ✉ georgeyouhana2@gmail.com | 📞 +20 122 335 0814 | 🌐 github.com/Geo-y20

🌐 linkedin.com/in/george-youhana-a5b756155 | 🌐 [Portfolio Link](#)

👤 Summary

AI Engineer with hands-on experience building and shipping production enterprise AI across three domains: **LLM systems & RAG** (bilingual chatbots, multi-provider assistants, Text-to-SQL), **computer vision & automation** (on-prem surveillance, OCR pipelines, ERP integration), and **analytics & infrastructure** (forecasting, Power BI, Dockerized deployments). Comfortable owning projects end-to-end — from architecture and modeling to API delivery and monitoring — with a track record of measurable impact inside a large enterprise group (EVA Group, 900+ users).

🎓 Education

Arab Academy for Science, Technology, and Maritime Transport, Alexandria, Egypt Feb 2021 – Feb 2025
Bachelor's Degree in Artificial Intelligence, Data Science Department.

- **Relevant Coursework:** Web Development, Data Structures, Machine Learning, Statistics for Data Science, Data Mining, Data Analysis, Data Science, Deep Learning, Natural Language Processing (NLP)

🏢 Work Experience

AI Engineer, Akhnaton Trading and Distribution – EVA Group Limited May 2025 – Present

LLM & AI Systems

- **Enterprise Sales Intelligence Platform:** Designed and deployed a **production-grade Text-to-SQL engine** over **Oracle Autonomous Database (10+ tables)**, enabling **natural language querying** with ~10s response latency.
- **Enterprise AI Assistant (ATR GPT):** Architected an **Azure-based RAG system** serving **900+ employees**, consolidating **20+ cross-department knowledge sources** within Microsoft Teams. Extended the platform with a **custom-built web interface** supporting an **assistant mode** with **multi-provider LLM routing** — allowing users to query through **OpenAI, Claude (Anthropic), and local Ollama-hosted models** via a unified API layer, enabling flexible model selection based on cost, privacy, and task requirements.
- **Eva Cosmetics HR Assistant (Offline Bilingual Chatbot):** Designed and deployed a **fully offline, on-prem bilingual (Arabic/English) HR policy assistant** for EVA Cosmetics, enabling employees to query internal HR policies via natural language. Built a **hybrid RAG pipeline** combining **ChromaDB vector search** with **BM25 keyword retrieval** for high-recall document grounding, powered by **Aya-Expansive 8B** as the local LLM — ensuring **zero data exposure** with no cloud dependency.
- **Conversational Data Analytics (Sheet Chat):** Built an **agentic chat-with-your-data feature** enabling users to query Excel/CSV files via natural language. Designed a **LangGraph-based multi-node pipeline** with **conditional routing** across intent-classified execution paths (SQL via DuckDB, visualization, report generation, data-quality analysis, semantic search), with a shared **typed state** flowing across nodes and an **LLM-powered explanation layer** converting raw results into natural language insights.

Vision & Automation

- **Security & Vision Intelligence Platform:** Designed and deployed an **on-prem surveillance system** integrated with cameras, implementing **fire/violence detection (~95%)**, **license plate recognition (~90%)**, driver ID scanning, face-recognition attendance, people counting, and gender detection with **alerting** across **CPU/GPU environments**.
- **OCR & Document Intelligence Platform:**
 - **Finance OCR:** Automated **cheque reading and validation workflows** before ERP ingestion.
 - **Supply Chain OCR:** Automated extraction of batch numbers, expiry dates, and manufacturing details using **Google Vision-based pipelines** for Arabic/English labels.

Analytics & Infrastructure

- **Forecasting & Decision Intelligence:** Built **production forecasting pipelines** using **time-series modeling (ARIMA, decomposition)** with evaluation and AI-generated explanations for supply chain planning.
- **Marketing & Social Listening Intelligence Platform:** Built a **cross-platform social listening system** analyzing **Arabic Facebook comments** using **sentiment analysis and text classification**, categorizing interactions into positive, negative, neutral, off-topic, price inquiries, FAQ, and delivery issues.
- **Supply Chain Analytics & BI Reporting:** Designed **Power BI dashboards** covering inventory turnover, inventory health, fulfillment efficiency, and order cycles, delivering **decision-support reporting** for supply chain operations.
- **Infrastructure & Deployment:** **Containerized AI services using Docker** with structured logging, monitoring, and optimized inference pipelines.

Data Science Freelance Coach, Digital Egypt Pioneers | Elharefa · Contract

Oct 2024 – Feb 2025

- Mentored two 8-week batches of up to 25 students each on freelancing, AI, and Data Science career readiness.
- Delivered 25+ hours per week of online mentorship, guiding technical and professional development.

Course Instructor, iSchool · Contract

Cairo, Egypt · Remote

Nov 2024 – Jan 2025

- Delivered hands-on coding lessons for Grades 4–6 using interactive problem-solving, projects, and visual learning.
- Designed custom curricula, monitored student performance, and maintained high levels of active engagement throughout the programme.

Machine Learning Engineer (Contract), GuruHub

Apr 2024 – Jul 2024

- Designed AI learning roadmaps and implemented ML-driven platform features to enhance user engagement.

AI Developer Intern, Elegant Textile Engineering (A&M Programming)

Jul 2023 – Aug 2023

- Applied machine learning techniques to optimize textile production workflows, achieving 94% model accuracy.
- Collaborated within a 4-member agile team and contributed to deployment of AI-driven production solutions.

 Projects

AI-Powered Learning Automation Platform

GitHub

Sep 2023 – Jul 2024

- Designed and deployed a FastAPI-based AI platform for automated transcription and content summarization of multimedia learning materials.
- Integrated OpenAI Whisper for speech-to-text processing and NLP pipelines for PDF content extraction and summarization.
- Processed 5,000+ files with ~95% transcription accuracy, reducing manual content review time by 50%.
- Implemented scalable API architecture supporting asynchronous processing workflows.

 Technical Skills

- **Core:** Python, SQL, FastAPI
- **AI & LLMs:** Large Language Models (LLMs), RAG, Prompt Engineering, Agentic Workflows, LangGraph, OpenAI APIs, Anthropic Claude API, Ollama (Local LLM Serving)
- **Machine Learning:** Scikit-learn, TensorFlow, Keras
- **Computer Vision:** OpenCV, YOLO, OCR Pipelines
- **Data & BI:** Power BI, Data Warehousing, KPI Analytics
- **Databases:** Oracle Autonomous DB, MySQL, MongoDB
- **Cloud & DevOps:** Microsoft Azure, Docker, CI/CD (GitHub Actions)